UrbanNest Hospitality Analytics Capstone Project

Group 6 – Cohort 6 Data Science Bootcamp

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Abstract

This study explores the key factors influencing hotel booking behavior and revenue performance using the Hotel Booking Demand dataset containing booking records from City Hotels and Resort Hotels between 2015 and 2017. The analysis combined exploratory data analysis (EDA) and machine learning techniques to identify major booking patterns, customer behaviours, and revenue drivers within the hospitality sector. The findings reveal that the booking environment is highly flexible and strongly influenced by online distribution channels, with transient two-adult travelers representing the dominant customer segment. City Hotels accounted for most bookings and revenue, while Resort Hotels were more closely associated with longer stays and leisure-oriented travel behaviour. Seasonal analysis showed that demand peaks during Summer, with early bookings contributing significantly to occupancy and revenue generation. Correlation and modelling analysis identified room nights, Average Daily Rate (ADR), customer type, and booking behaviour as the strongest predictors of revenue performance. A dual modelling approach was adopted using both Revenue and ADR models to separate occupancy-driven and pricing-driven performance. The Revenue model achieved a high R^2 score of 0.86, primarily driven by room nights and stay duration, reflecting strong volume-based effects. In contrast, the ADR model achieved a moderate R^2 score of 0.52, indicating that pricing behaviour is influenced by more complex factors such as room type, seasonality, and booking channels. The study also highlighted the importance of stay duration and pricing strategy in hotel profitability. While Online Travel Agents (OTAs) dominated booking channels, opportunities were identified to strengthen direct bookings, improve customer retention, and expand higher-value segments such as family, group, and corporate travelers. The findings demonstrate that effective hotel revenue optimisation requires a combination of dynamic pricing, targeted customer segmentation, seasonal demand management, and strategies that encourage longer stays and higher-value bookings. The study further recommends the use of advanced machine learning models such as Random Forest and Gradient Boosting techniques for future predictive modelling and revenue forecasting within the hospitality industry.

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1. Introduction

1.1 Business and Industry background

The hospitality industry is highly competitive, with hotels constantly trying to balance pricing, occupancy, and customer satisfaction. Revenue management is crucial, as small changes in booking patterns, pricing strategies, and customer preferences can significantly impact overall profitability. Traditionally, many hotels rely on historical assumptions rather than data-driven insights, leading to inefficiencies in decision-making. This often leads to missed revenue opportunities during peak periods and underperformance during low-demand seasons. However, Modern revenue management systems now rely heavily on data to forecast demand, adjust pricing dynamically, and optimize occupancy rates [1], [2]. By applying data analysis and machine learning techniques, hotels can better understand customer behaviour and improve revenue forecasting, which directly impacts profitability and operational efficiency. In the hospitality industry, the Hotel Booking Demand Dataset [3] provides real-world booking data containing over 100,000 observations from both city and resort hotels, including features such as stay duration, number of guests, and booking lead time. These features are important to understand booking patterns, customer behaviour, and revenue performance. In this work, UrbanNest Hospitality Analytics aims to apply data science techniques to better understand booking behaviour and improve revenue forecasting using the dataset. A Linear Regression model is developed to predict revenue drivers based on booking characteristics such as guest count, room type, lead time, and seasonal factors.

1.2 Problem statement

Hotel management often faces challenges in accurately forecasting revenue due to variations in customer behaviour, booking channels, seasonal demand, and room allocation strategies. Without reliable forecasting tools, hotels may:

- Underprice rooms during high-demand periods.
- Overestimate expected revenue.
- Allocate resources inefficiently.
- Miss opportunities for revenue optimisation.

The business challenge is therefore to develop a predictive framework capable of estimating hotel revenue and identifying the primary drivers of revenue generation.

2. Aim and Objectives

The aim of this project is to develop a linear regression model capable of predicting hotel revenue based on booking related features. The objectives are to:

- explore and understand the hotel booking data
- identify key factors influencing revenue
- perform data cleaning and preprocessing
- conduct exploratory data analysis (EDA)
- perform basic SQL-based data analysis
- build a Linear Regression model for revenue prediction
- evaluate the performance of the model using standard machine learning metrics.
- provide actionable business recommendations

3 Data collection, understanding, cleaning and preparation

3.1 Data collection and understanding

The dataset used in this project is the publicly available Hotel Booking Demand dataset, which contains booking information from both City Hotels and Resort Hotels between 2015 and 2017 [3]. The dataset was obtained from Kaggle [4], an online repository widely used for sharing datasets and machine learning resources. The data was subsequently imported into the analysis environment using the Pandas library in Python. The original dataset consisted of 119,390 observations and 32 variables. The key variables and their corresponding data types are presented in **Figure 1.** Initial data inspection revealed the presence of missing values in the country, children, agent, and company columns, indicating the need for data cleaning and preprocessing before analysis and model development.

In addition, the children variable was initially stored as a float data type. Since the number of children should only take whole-number values, the variable was converted to an integer (int64) format to ensure consistency and validity within the dataset. Preliminary exploration of the numerical variables also revealed potential positive skewness within the ADR (Average Daily Rate) column. The ADR variable contains extreme values, with a maximum value of 5400 and a minimum value of -6.38. These unusual values suggest the presence of outliers and possible data quality issues, which required further investigation and preprocessing prior to modelling.

	Column	Non-Null Count	Missing Values	Data Type
24	company	6797	112593	float64
23	agent	103050	16340	float64
13	country	118902	488	str
10	children	119386	4	float64
4	arrival_date_month	119390	0	str
5	arrival_date_week_number	119390	0	int64
0	hotel	119390	0	str
1	is_canceled	119390	0	int64
7	stays_in_weekend_nights	119390	0	int64
6	arrival_date_day_of_month	119390	0	int64
9	adults	119390	0	int64
8	stays_in_week_nights	119390	0	int64
11	babies	119390	0	int64
12	meal	119390	0	str
2	lead_time	119390	0	int64
3	arrival_date_year	119390	0	int64
15	distribution_channel	119390	0	str
14	market_segment	119390	0	str
18	previous_bookings_not_canceled	119390	0	int64
16	is_repeated_guest	119390	0	int64
19	reserved_room_type	119390	0	str
20	assigned_room_type	119390	0	str
21	booking_changes	119390	0	int64
17	previous_cancellations	119390	0	int64
22	deposit_type	119390	0	str
25	days_in_waiting_list	119390	0	int64
26	customer_type	119390	0	str
27	adr	119390	0	float64
28	required_car_parking_spaces	119390	0	int64
29	total_of_special_requests	119390	0	int64
30	reservation_status	119390	0	str
31	reservation_status_date	119390	0	str

Figure 1: *Overview of the dataset structure, including each variable, its data type, number of non-null observations, and corresponding missing values.*

3.2 Data Cleaning and Preparation

3.2.1 Missing value treatment

As shown in **Figure 1**, four variables contained missing observations. These missing values were addressed using a context-driven approach informed by the meaning and expected behaviour of each variable. For the children related fields, missing entries were interpreted as indicating bookings without children, as it is common in booking systems for such values to be left blank rather than explicitly recorded as zero. Considering this, these missing values were imputed with zero to preserve consistency and avoid overstating missingness.

In the case of the country variable, missing values were replaced with "Unknown" to retain these records within the dataset while clearly distinguishing them from valid country entries. This approach helps prevent unnecessary data loss while maintaining transparency about incomplete information.

Furthermore, the agent column was removed due to its limited analytical value within the scope of this study Although the column contained valid entries, it primarily functioned as an identification code for booking agents and did not provide meaningful explanatory value for the analysis objectives as there. Similarly, the company column was excluded due to a substantial proportion of missing data, which could introduce bias or reduce the reliability of any insights derived from it. Retaining such a sparsely populated variable would add noise rather than meaningful information to the analysis.

3.2.2 Data type correction

Variables were converted into appropriate numerical and categorical data formats to ensure compatibility with subsequent analysis and modelling processes. Specifically, the children column was converted from float to int64, while the reservation_status_date column was transformed from string format to a datetime data type to support time-based analysis and feature extraction.

3.2.3 Duplicate handling

Approximately 32,020 duplicate observations were initially identified in the dataset. However, upon closer examination, these records were determined to be valid and contextually appropriate, particularly given the nature of hospitality data. In the hospitality industry, apparent duplicates frequently occur due to group bookings, shared reservation dates, and multiple room allocations under a single booking reference. For instance, guests within the same group may check in on the same date but be assigned different rooms, resulting in records that appear duplicated when viewed superficially but represent distinct transactions or service units. Removing these 32,020 observations would compromise the integrity and representativeness of the dataset. Such deletion could distort occupancy and booking patterns which could underrepresent group booking behavior and lead to bias downstream analytical models. Therefore, these records were retained as their inclusion ensures the dataset remains comprehensive, accurate, and reflective of real-world booking scenarios.

3.2.4 Outlier Treatment

Extreme values were observed in several variables, particularly Lead Time and Average Daily Rate (ADR). While these outliers warranted investigation, only the ADR variable required adjustment due to the presence of negative values and unusually high extremes, both of which could significantly distort derived metrics such as revenue. Initially, ADR ranged from -6.38 to 5400. Removing these observations could result in the loss of valuable information, especially from high-value customer segments. So therefore, a capping (winsorization) approach [5] was applied. This method preserves the overall distribution of the data while minimizing the influence of extreme values. ADR values were capped using quantile thresholds at the 0.001 (minimum) and 0.999 (maximum) levels. Further investigation into this shows that only one observation showed an ADR of 5400 for a one-night stay in a standard room category booked through an offline travel agency. This value was considered inconsistent with the general pricing structure of the dataset and treated as an unrealistic outlier. After applying the capping procedure, the ADR range was adjusted to 0-326, which aligns more closely with realistic pricing structures observed in the hospitality industry. The minimum value of zero likely represents complimentary stays, promotional bookings, or staff accommodations, all of which are valid operational scenarios and therefore retained. As shown in **Figure 2**, the adjusted ADR distribution is now relatively balanced and analytically useful. Most ADR values fall between 67.5 and 125, with a mean of 99.7 and a median of 92.4. It is notable that the distribution is moderately right-skewed, and this is typical of hotel pricing data [6]. Values exceeding 125 likely correspond to premium or luxury bookings. Following these adjustments, the ADR variable exhibits a standard deviation of 48.47, indicating a reasonable level of variability while avoiding distortion from extreme outliers.

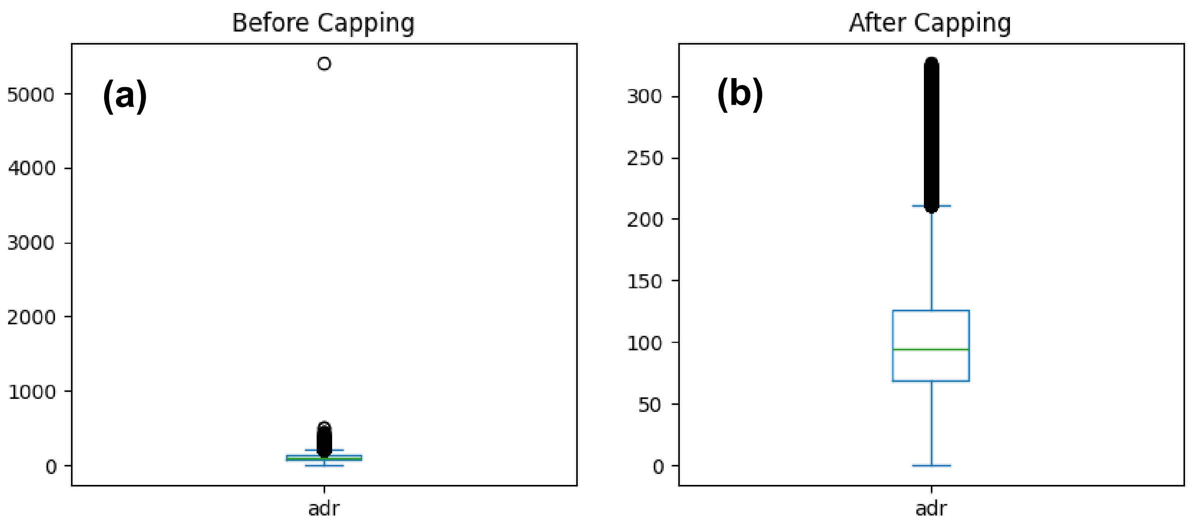

Figure 2(a) *Distribution of Average Daily Rate (ADR) before capping, showing the presence of negative values and extreme outliers;* ***(b)*** *Distribution of ADR after capping* *at the 0.001–0.999 quantile thresholds, illustrating a more realistic and well-spread range of values aligned with typical hospitality pricing.*

3.2.5 Feature Engineering

To enhance both predictive performance and business interpretability, several features were engineered from the existing variables. These transformations were designed to improve data usability, capture meaningful customer behavior, and support more insightful analysis.

Booking Identification

To improve record traceability and dataset organization, a Booking ID feature was created. This unique identifier was positioned as the first column in the dataset to facilitate easy referencing, indexing, and management of individual booking records.

Guest Composition

A new feature (total_guests) was derived by aggregating the number of adults, children, and babies associated with each booking. This provides a clearer representation of party size and overall occupancy. In addition, guest composition was further categorized into meaningful segments to support behavioural and demographic analysis:

- **single:** 1 adult, 0 children, 0 babies
- **two_adults:** 2 adults, 0 children, 0 babies
- **group:** More than 2 adults, 0 children, 0 babies
- **family:** At least one child or baby present

Booking Behaviour

The lead_time variable was transformed into a categorical feature (lead_category) based on established industry patterns (as referenced by OnlineHotelInsight [7]). This categorization is outlined in **Table 1**. The objective of this transformation is to distinguish between different booking behaviours where last-minute bookings typically reflect urgent or spontaneous travel decisions, short-term bookings often associated with leisure travel or domestic trips, medium-term bookings are indicative of moderately planned travel. Early bookings are characteristics of long-term planners, including international travellers and group reservations. By structuring booking timelines into these categories, the dataset better captures customer intent and planning behaviour, which is critical for forecasting, demand modelling, and targeted marketing strategies.

Table 1: *Classification of bookings into lead categories based on lead time.*

Lead time (days)	Booking categorisation
0 -7	Last-minute
7-30	Short-term
30-90	Medium-term
>90	Early-booking

Room Features

The dataset includes both reserved_room_type and assigned_room_type, which provide insight into customer expectations versus actual allocation. To capture discrepancies between the two, a new binary feature (room_change) was engineered to indicate whether a booking experienced a room type change. This variable is valuable for analyzing operational efficiency and its potential impact on customer satisfaction.

Stay Features

A new feature (room_nights) was created to represent the total duration of each stay. This was derived by aggregating stays_in_weekend_nights and stays_in_weekday_ nights. The Room Nights variable serves as a critical input for revenue estimation, as it reflects the full length of a guest's stay.

Revenue

The primary target variable in this study is Revenue, and it was engineered by combining pricing and stay duration. Specifically, revenue was calculated as the product of Average Daily Rate (ADR) (i.e., the price per room per night) and Room Nights, as shown in **Equation 1.**

$$Revenue = ADR \times room\_nights \quad \mathbf{1}$$

Seasonal Features

To better understand the influence of seasonality on booking patterns and revenue generation, the variables arrival_date_year, arrival_date_month, and arrival_date_day_of_month were consolidated into a single date feature (arrival_date). From this unified variable, multiple timing-related features were derived, including month (numeric and name formats), weekday, and weekend indicator (is_weekend). These features enable more granular analysis of booking trends, such as identifying periods of higher demand or increased willingness to pay. For improved interpretability, months were further grouped into seasonal categories using standard mappings as shown below:

- Winter: December, January, February
- Spring: March, April, May
- Summer: June, July, August

- Autumn: September, October, November

This seasonal classification is aimed to provide clearer insights into how different times of the year influence customer behaviour and revenue performance.

Country Feature

To better evaluate the geographical distribution of bookings and revenue contribution, the country variable originally recorded as abbreviated codes was transformed into full country names, resulting in a new feature (country_full). This transformation was performed using the pycountry library. Entries that did not conform to standard codes or contained missing values were categorised under 'Unknown' label to maintain consistency.

3.2.6 Further Data Quality Check

A further data quality assessment was conducted prior to proceeding to the exploratory data analysis (EDA) phase to ensure the integrity and analytical relevance of the dataset. Given that the primary objective of this study is to investigate the drivers of actual revenue, the analysis was restricted to confirmed (non-cancelled) bookings only. The original dataset contained approximately 44,224 cancelled bookings, representing a substantial proportion of the data. While such records are valuable for analysing cancellation behaviour and identifying segments prone to booking cancellations, they do not contribute to realized revenue. Consequently, all cancelled bookings were excluded so that the dataset reflects only revenue-generating transactions.

In addition, several data inconsistencies and anomalies were identified and addressed to improve overall data quality:

- 31 bookings with recorded revenue but zero guests were removed, as these entries are logically inconsistent.
- 294 records containing children and/or babies without any adults were excluded, given that hotels do not accommodate minors without adult supervision in reality.
- 1,622 bookings with registered guests but zero revenue were also dropped. While some of these may represent complimentary bookings, the majority were associated with zero-night stays indicates invalid or incomplete records.
- 2 extreme outliers involving highly unrealistic occupancy levels (e.g., 10 children and 9 babies in a single booking) were removed, as such scenarios are likely not possible in real-world hospitality operations.

Following these preprocessing steps, the dataset was refined to 73,167 observations and 45 features for subsequent exploratory data analysis (EDA).

4 Exploratory Data Analysis (EDA)

The EDA combines univariate, bivariate, and multivariate analysis to better understand booking behaviour and the major driver of revenue.

4.1 Booking characteristics

The booking characteristics of the dataset shown in **Figure 3** provide insight into who is booking, the booking context, the hotel type selected, and the channels through which bookings are made. The dataset consists of bookings from two hotel categories (City Hotel and Resort Hotel). As shown in **Figure 3a**, the City Hotel accounts for a substantially larger proportion of bookings, representing approximately 61.5% more bookings than the Resort Hotel. This difference may be influenced by several factors, including location, greater accessibility, competitive pricing, and increased booking flexibility compared to resort hotel.

The distribution of bookings across different guest types is shown in **Figure 3b**. The results indicate that most bookings are made by the two-adult category, which may represent either couples or two individuals sharing a room. This segment clearly dominates the booking composition. The family segment accounts for approximately 7.9% of total bookings, while the group category defined as bookings with more than two adults and no children or babies represents the smallest share. However, it is important to note that this "group" classification may not fully capture actual group travel behaviour, as larger groups often split their bookings across multiple rooms, thereby being reflected as separate smaller bookings in the dataset. Given this limitation, a more practical segmentation may be to classify bookings into family versus non-family categories.

The distribution of bookings across customer types as shown in **Figure 3c** revealed that most bookings are generated by transient customers (70.6%), followed by transient-party bookings (24.9%), indicating a strong reliance on individual and small-group travellers. In contrast, contract (3.8%) and group bookings (0.7%) contribute minimally to overall demand. This pattern suggests that the hotel primarily operates in a flexible, short-term booking environment, with limited engagement in long-term contracts or large group reservations. While this provides opportunities for dynamic pricing, it also highlights potential areas for strategic expansion into more stable and high-volume customer segments.

Figure 3d shows the distribution of bookings across market segments. The results indicate that nearly half of all bookings (48.1%) are generated through Online Travel Agents (OTAs), followed by Offline Travel Agents and Tour Operators(offline-TA/TO) (21.3%). Direct bookings account for 14.2%, while group and corporate segments contribute more modest shares of 10.2% and 5.8%, respectively. This simply means the hotel strongly depend on intermediary channels, with over two-thirds of bookings originating from external agents. Although this enhances market reach, it may reduce profit margins due to commission costs. In contrast, the relatively low proportion of direct bookings suggests an opportunity for hotels to strengthen their direct sales channels and improve revenue optimisation.

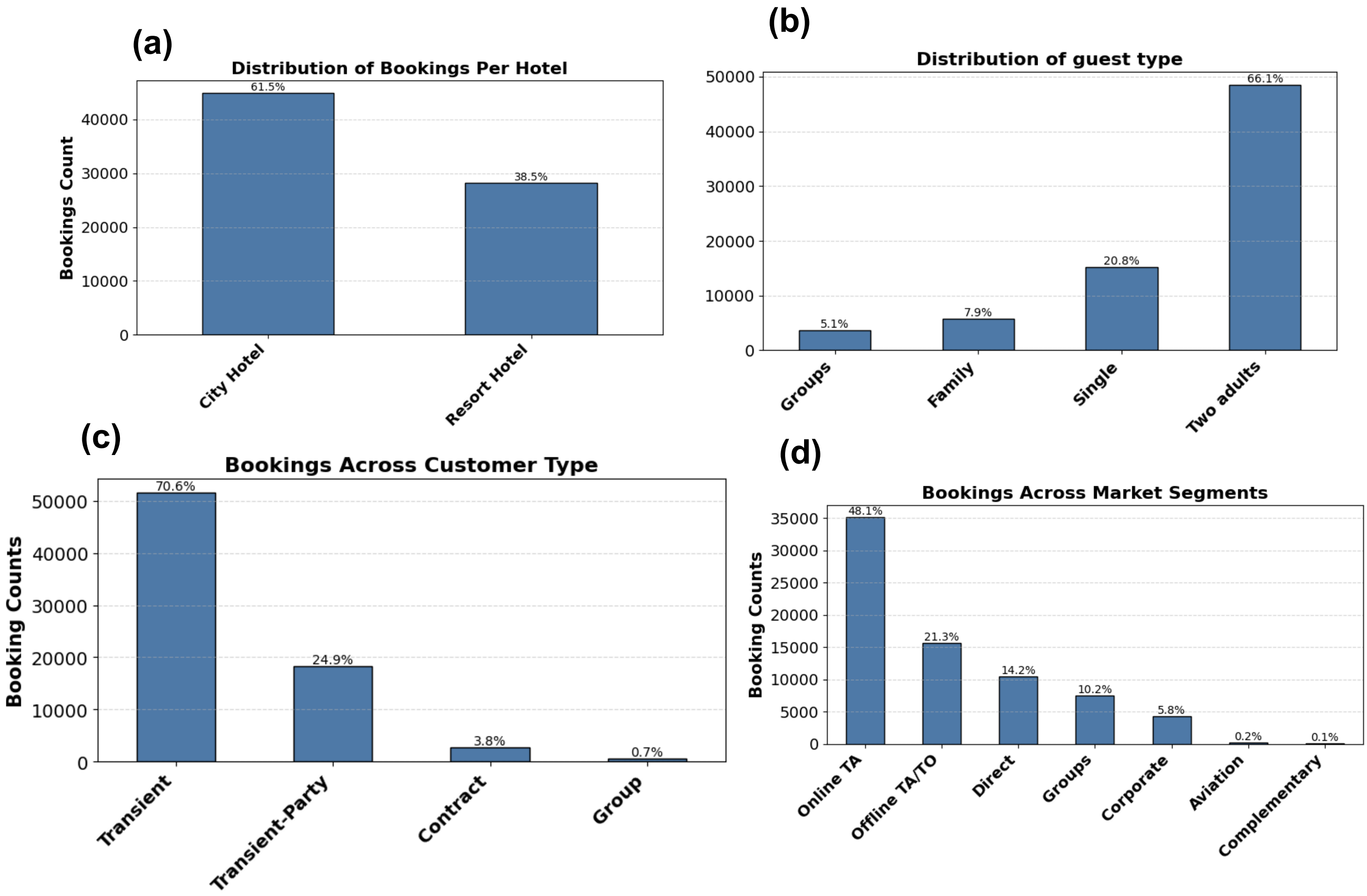

Figure 3: *Analysis of Booking Characteristics Showing Hotel Type Preferences, Guest Categories, Customer Types, and Market Segmentation Patterns*

4.2 Customer Booking behaviour

Figure 4 shows customer booking behaviour by combining key factors such as lead time, hotel type, guest composition, market segment, and customer type. The idea is to better understand how customers book hotels, identify the main patterns driving demand, and highlight opportunities that could improve hotel revenue and customer targeting. As shown in **Figure 4a**, most bookings are made in advance with early bookings dominating the reservations across both City Hotels and Resort Hotels. This suggests that a large proportion of customers prefer to plan their trips ahead of time rather than booking at the last minute. However, City Hotels consistently record higher booking volumes across all lead-time categories, especially within early and medium-term bookings. This pattern suggest that City hotel is likely often used by business travellers and short-stay visitors who travel more frequently and follow relatively predictable booking patterns. In comparison, Resort Hotels show stronger participation from both early planners and last-minute travellers. This may reflect the nature of leisure travel where guests tend to plan vacations in advance while others make spontaneous travel decisions. It is likely that Resort Hotels are associated with longer stays and more seasonal or variable demand, whereas City Hotels operate within a more stable, high-frequency booking environment.

When guest composition is considered (**Figure 4b**), bookings involving two adults clearly dominate across all lead-time categories. This group likely represents couples or two individuals travelling together and forms the largest source of demand within the dataset. Their strong presence in early and medium-term bookings further suggests that this customer group tends to organise trips ahead of time. On the other hand, single travellers appear more frequently in last-minute bookings, which may reflect urgent or business-related travel. Although family and group bookings represent only a small share of total demand, these customers are likely to pay more, stay longer and potentially generate higher overall revenue per booking. Their low representation indicates an important imbalance within the booking structure, while most bookings come from smaller travel parties, larger and potentially more profitable segments remain relatively underdeveloped.

As shown in **Figure 4c**, travellers strongly depend on intermediary booking channels, particularly OTAs which dominate bookings across all lead-time categories. This suggests that OTAs play an important role in both planned and last-minute reservations, showing how heavily customers rely on online platforms when searching for accommodation and comparing prices. Offline-TA/TO also contribute meaningfully especially for early and medium-term bookings, which may be linked to organised travel arrangements and package holidays. In contrast, direct bookings, corporate bookings, and group segments account for much smaller proportions of demand. This confirms earlier findings that the hotel relies heavily on external distribution channels, with nearly half of all reservations coming through OTAs. While these channels increase visibility and help hotels reach wider markets, they likely reduce profit margins through commission costs and limit direct relationships with customers.

The customer type analysis in **Figure 4d** further strengthens the view that the booking environment is heavily driven by flexible and short-term travel demand. Transient customers account for most bookings across all lead-time categories, particularly within early and medium-term reservations. This suggests that many customers plan ahead but still prefer flexibility in their travel arrangements. Similarly, transient-party customers who represent small groups of independent travellers, also contribute significantly to total demand. By comparison, contract and group customers contribute only a very small share of bookings, indicating limited reliance on long-term agreements and organised large-scale reservations. This pattern reinforces earlier findings that more than 95% of bookings are generated by transient and transient-party customers, indicating the hotel's dependence on flexible individual travel demand rather than stable contractual business.

Considering the results, a clear picture of the typical customer begins to emerge. The dominant customer is generally a transient traveller booking for two adults, making reservations through OTAs and planning trips in advance through early or medium-term bookings. Demand is concentrated more heavily within City Hotels, while Resort Hotels attract more leisure-focused and variable travel behaviour.

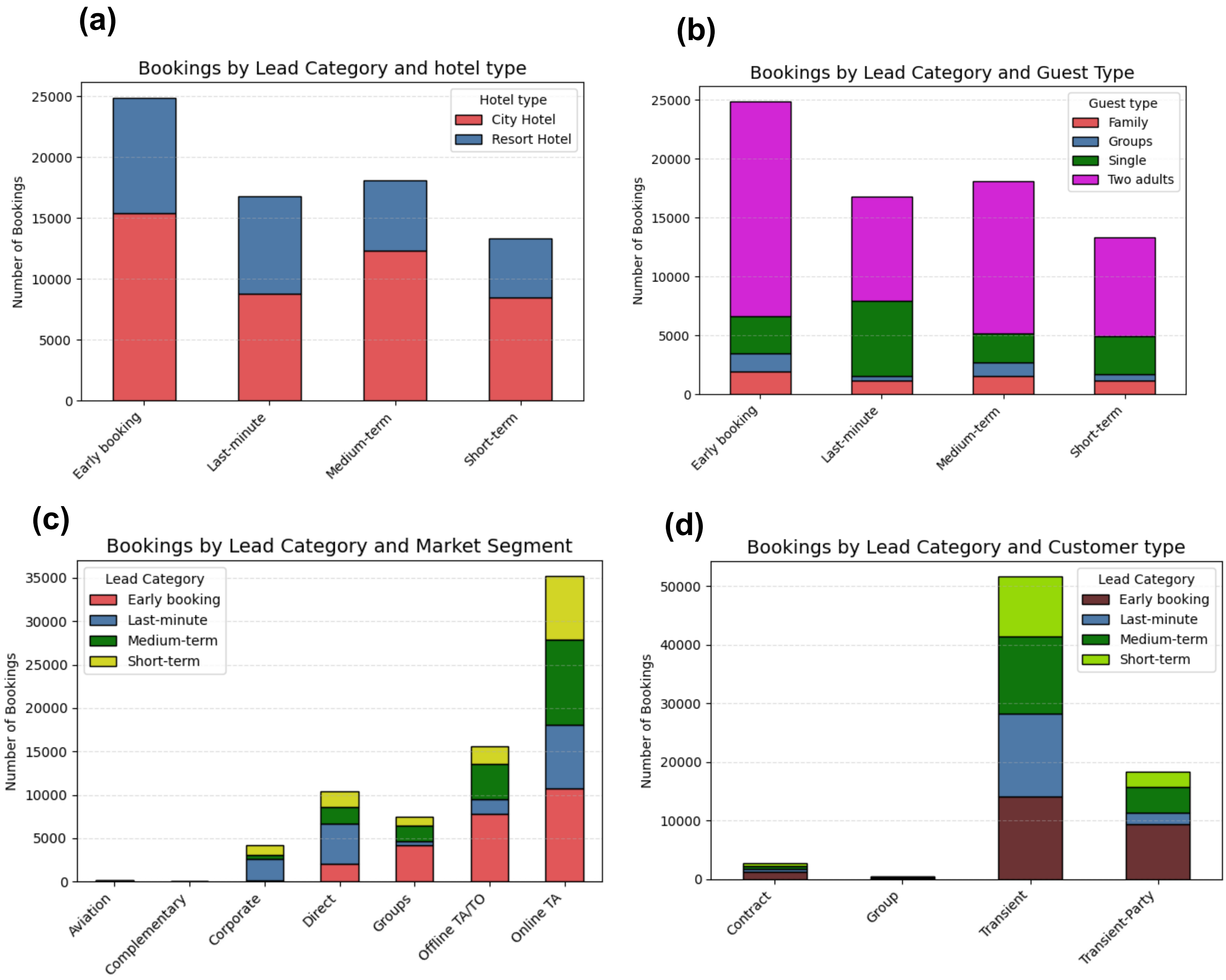

Figure 4: *Distribution of bookings by lead time, showing variations across (a) hotel type, (b) guest type, (c) market segment, and (d) customer type.*

4.3 Pricing and Revenue Analysis

The ADR distribution in **Figure 5a** shows a positively skewed pattern, with most bookings concentrated within the moderate price range. Most reservations fall between approximately 50 and 150 ADR with the highest concentration occurring around the mid-range pricing levels. This suggests that the hotels primarily operate within an affordable to moderately priced market segment. This tends to attract customers who are price-conscious while still willing to pay for comfort and convenience. As ADR values increase beyond the mid-range categories, the number of bookings gradually declines. Only a relatively small proportion of bookings are associated with very high ADR values above 200. This suggests that premium or luxury-priced reservations are less common within the dataset. This pattern reflects a customer base that is largely driven by value-oriented travel behaviour rather than exclusive high-end accommodation demand.

The revenue distribution also demonstrates the same strong positive skew (**Figure 5b**), where most bookings generate relatively low to moderate revenue values, while a small number of bookings contribute exceptionally high revenue amounts. Most bookings are concentrated within the lower revenue range, indicating that high booking frequency is most likely driven by shorter stays and moderate room rates. However, the long tail observed within the revenue distribution suggests the presence of a smaller segment of high-value bookings that generate significantly greater revenue. These high-revenue bookings are likely associated with customers who stay for longer durations, select premium room options, or combine higher ADR values with extended stays. Although these bookings occur less frequently, they contribute disproportionately to total revenue and therefore represent an important source of profitability.

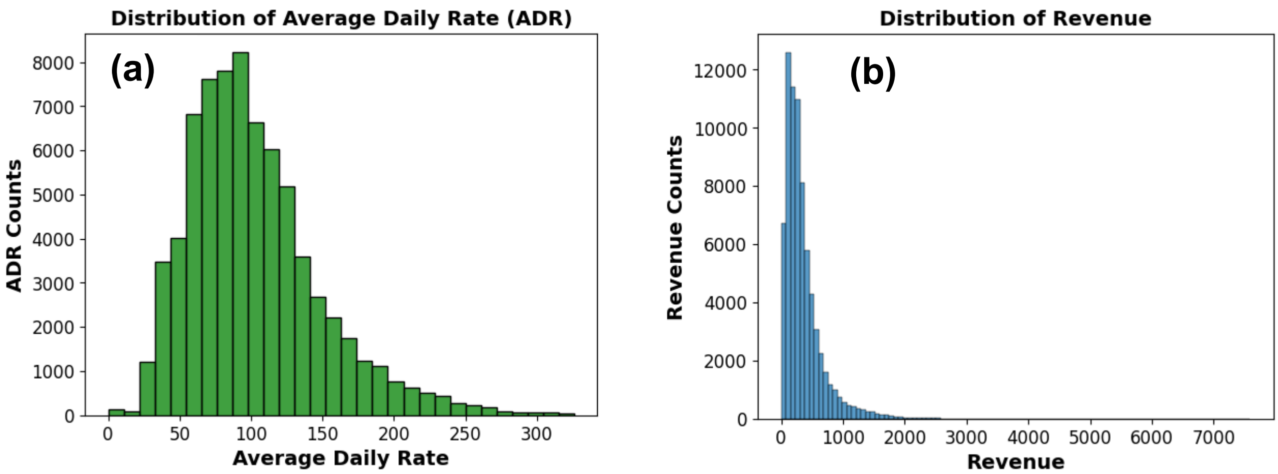

Figure 5: *Distribution Patterns of* ***(a)*** *Average Daily Rate (ADR) and* ***(b)*** *Revenue Generated from Hotel Bookings.*

Figure 6 compares the average ADR and total revenue generated across different guest group types. The results show that family and group travellers record the highest average ADR (**Figure 6a**), both contributing approximately 31% each, indicating a greater willingness to pay higher room rates compared to other customer segments. In contrast, single travellers contribute the lowest average ADR, while two-adult bookings remain within the moderate pricing range.

Despite paying lower ADR on average, the two-adult segment generates the largest share of total revenue, accounting for approximately 67.6% of overall revenue(**Figure 6b**). This is primarily driven by the high volume of bookings from this segment, reinforcing earlier findings that two-adult transient travellers represent the dominant customer group within the dataset. Family, single, and group travellers contribute significantly smaller shares of total revenue despite their relatively higher ADR levels.

An important relationship between booking volume and pricing behaviour is now evident, while high-value segments such as families and groups pays more per booking, the hotel's overall revenue performance remains heavily dependent on the consistent and high-frequency demand generated by two-adult travellers. This suggests that family and group travellers represent important potential growth markets for hotels. Their willingness to pay higher rates, combined with increased overall spending, creates opportunities for hotels to develop targeted packages, larger accommodation options, and personalised services aimed at attracting these higher-value customer segments.

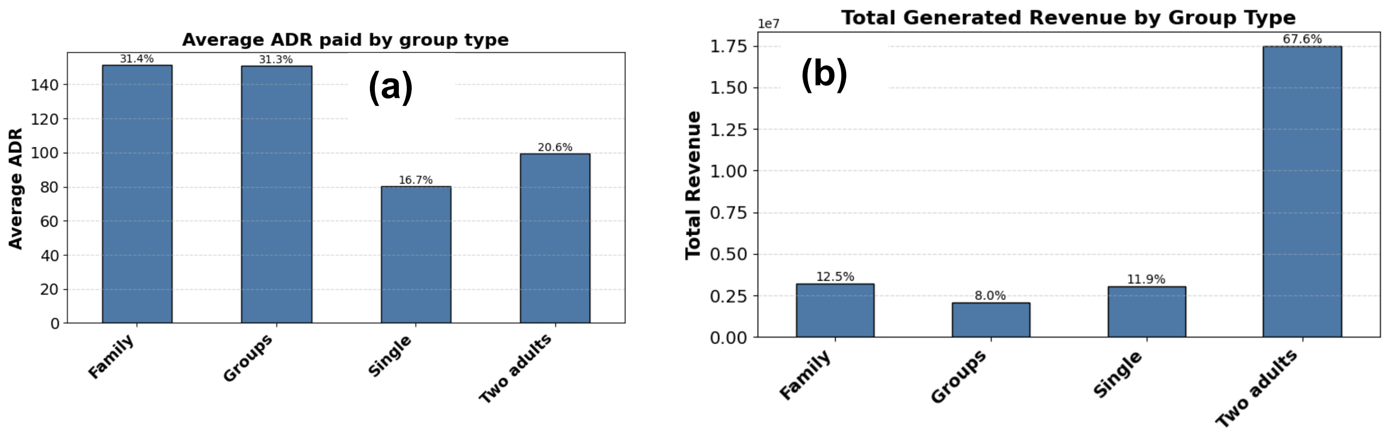

***Figure 6 (a)** Average Daily Rate (ADR) and **(b)** Total Revenue Contribution by Guest Group Type*

Figure 7 presents the relationship between Average Daily Rate (ADR) and total revenue across assigned room types. The ADR analysis in **Figure 7a** shows that Room Types G and H record the highest average ADR values, followed closely by Room Type F, indicating that these room categories are positioned as more premium accommodation options. In contrast, Room Types A, B, and I generate comparatively lower ADR values, suggesting that they are more affordable and likely targeted toward the hotel's mainstream customer base.

However, a different pattern emerges in total revenue generated by each room as observed in **Figure 7b**. Despite having relatively moderate ADR levels, Room Type A contributes the highest share of total revenue by a substantial margin, followed by Room Type D and Room Type E. This suggests that these room categories experience significantly higher booking volumes and occupancy rates, making them the hotel's primary revenue drivers. In comparison, premium-priced room categories such as G and H contribute far less to total revenue despite their higher ADR values. This indicates that although these rooms generate more revenue per booking, they attract fewer reservations overall. This is because the hotel business model remains strongly volume-driven, where consistent occupancy and high booking frequency generate more overall revenue than premium pricing alone.

However, the higher ADR associated with premium room types suggests the presence of smaller but higher-value customer segments willing to pay more for enhanced accommodation experiences. This creates an opportunity for hotels to strengthen premium offerings, luxury packages, and personalised services in order to increase the occupancy and profitability of higher-priced room categories.

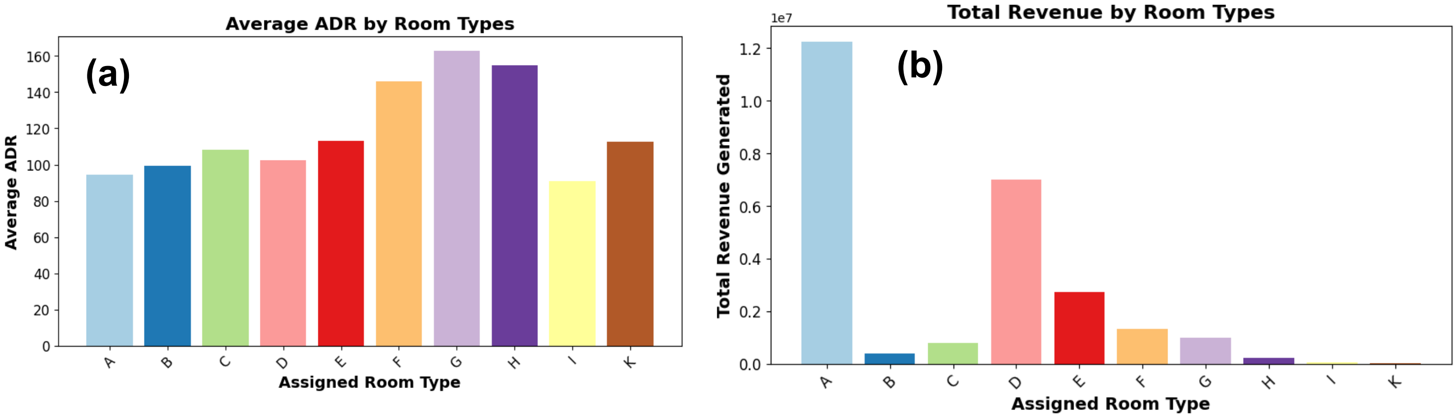

***Figure 7**: Comparison of **(a)** Average Daily Rate (ADR) and **(b)** Total Revenue Across Assigned Room Types*

The customer type analysis in **Figure 8a** shows that transient customers dominate both ADR and revenue generation, with the widest spread of data points and the highest concentration of high-revenue bookings. This further confirmed that transient travellers represent the hotel's primary customer base and are the main drivers of overall revenue performance. Transient-party customers also contribute meaningfully, although at a lower scale, while contract and group customers remain comparatively limited in both booking volume and revenue contribution.

From the market segment perspective (**Figure 8b**), there is a strong role of OTAs in revenue generation as their bookings account for a substantial proportion of reservations across a broad ADR range. Direct bookings also show several high-revenue observations, suggesting that although smaller in volume, direct channels may attract higher-value customers. Offline-TA/TO contribute moderately, while corporate, group, and aviation segments generate comparatively lower revenue concentrations.

The presence of several high-revenue outliers across transient and direct booking segments suggests that certain customers generate exceptionally high booking values likely due to longer stays, and premium room selections. This means that there are opportunities to increase profitability through higher-value direct bookings and premium customer targeting, even though the current revenue is strongly driven by high-volume transient customers.

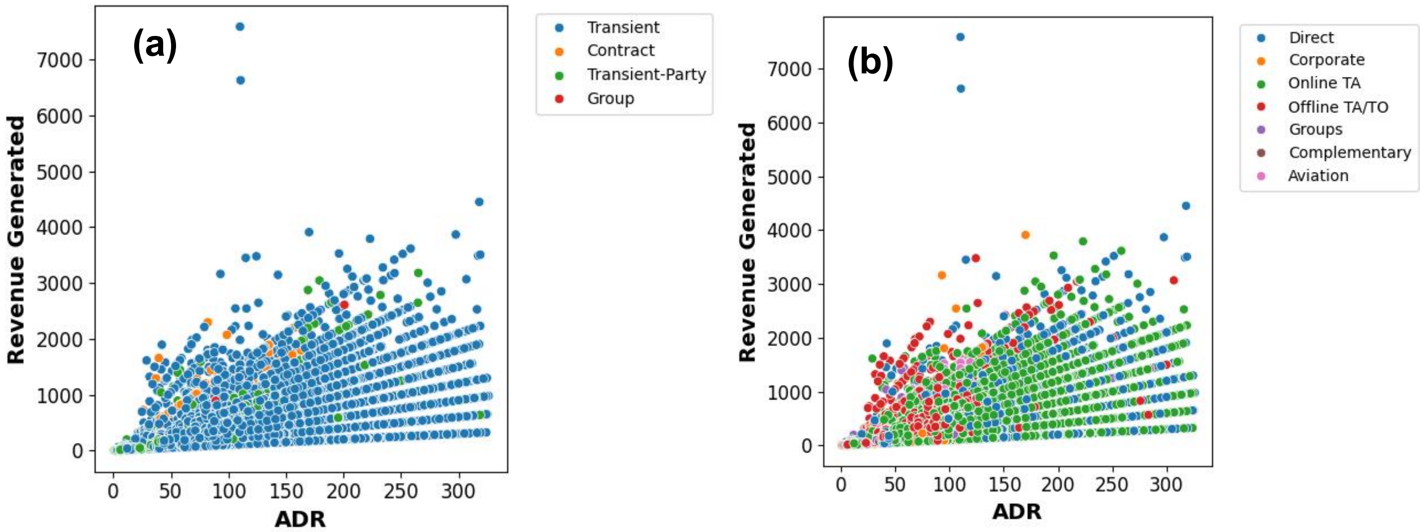

Figure 8: *Comparison of Average Daily Rate (ADR) and Total Revenue Across* ***(a)*** *customer type and* ***(b)*** *market segment*

4.4 Customer Stay Behaviour

Figure 9 presents the distribution of total nights stayed per booking and the contribution of different guest types to overall stay duration. The distribution of total nights stayed (**Figure 9a**) is heavily concentrated within shorter stays, with most bookings lasting between one and five nights. As the length of stay increases, the number of bookings declines significantly, indicating that long-stay reservations are relatively uncommon within the dataset. However, a small number of extended stays are observed via the long-tail distribution reflecting occasional high-duration bookings.

The analysis by guest type in **Figure 9b** further shows that two-adult travellers account for the largest share of total nights stayed, contributing approximately 70.7% of overall stay duration. This confirmed the earlier findings that two-adult transient travellers represent the hotel's dominant customer segment and are the primary drivers of occupancy, booking volume, and revenue generation. Single travellers contribute a moderate share of total nights stayed, while family and group segments account for comparatively smaller proportions.

Although family and group travellers contribute fewer overall nights, earlier analysis showed that these segments tend to pay higher ADR values and may generate greater revenue per booking. This suggests that while the hotel's current occupancy is largely sustained by high-frequency short stays from two-adult travellers, family and group customers remain valuable growth segments with strong potential for longer stays and higher-value bookings.

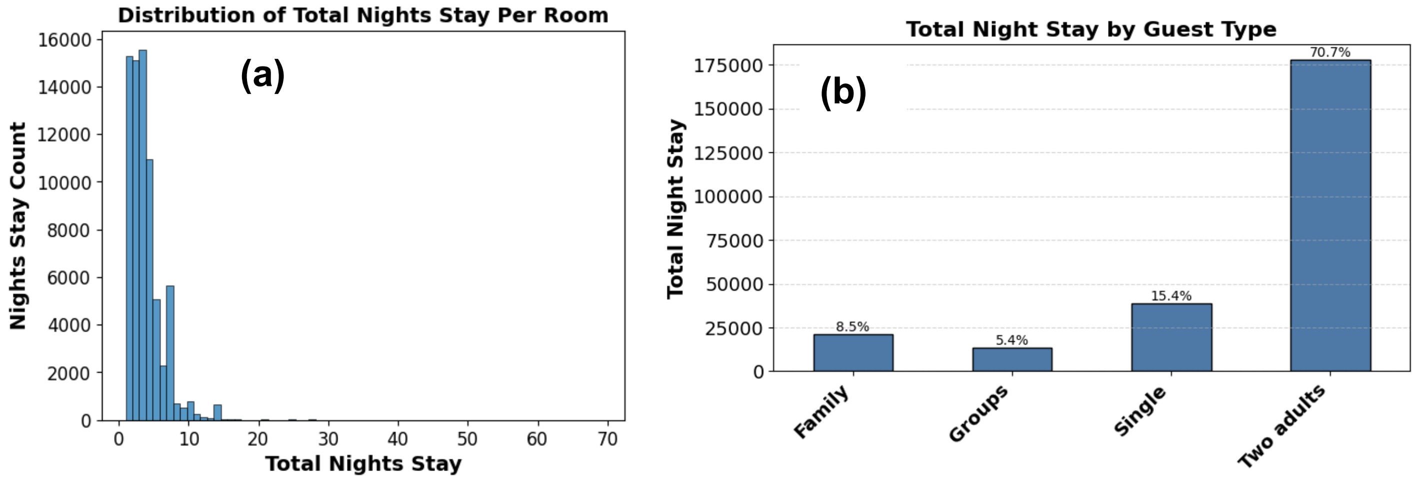

Figure 9: *The distribution of total night stay per **(a)** bookings, and **(b)** across different guest type*

The distribution of room nights across hotel types and the relationship between ADR, total room stay, and revenue generation is presented in **Figure 10**. The boxplot analysis in **Figure 10a** shows that Resort Hotels generally record longer stay durations compared to City Hotels, with a wider spread of room nights and several extreme outliers extending beyond 60 nights. In contrast, City Hotel stays are more concentrated around shorter durations, supporting the high-frequency and short-stay booking behaviour identified in earlier analyses. These findings confirmed earlier hypothesis that Resort Hotels are more associated with leisure-oriented and extended-stay travel behaviour, while City Hotels mainly serve transient and short-duration travellers.

As shown in the scatterplot analysis in **Figure 10b**, there is a strong positive relationship between ADR, total room nights, and revenue generation. Revenue consistently increases as both ADR and stay duration rise, indicating that higher room rates combined with longer stays generate substantially greater booking value. The concentration of points within lower-to-mid ADR ranges also supports earlier findings that the hotel primarily operates within a moderate pricing market supported by high booking volume.

Several high-revenue outliers are associated with both high ADR values and extended room stays shows that a smaller number of premium or long-duration bookings contribute disproportionately to total revenue. This aligns closely with previous analyses indicating that longer stays, higher ADR values, and premium customer segments represent important drivers of hotel profitability.

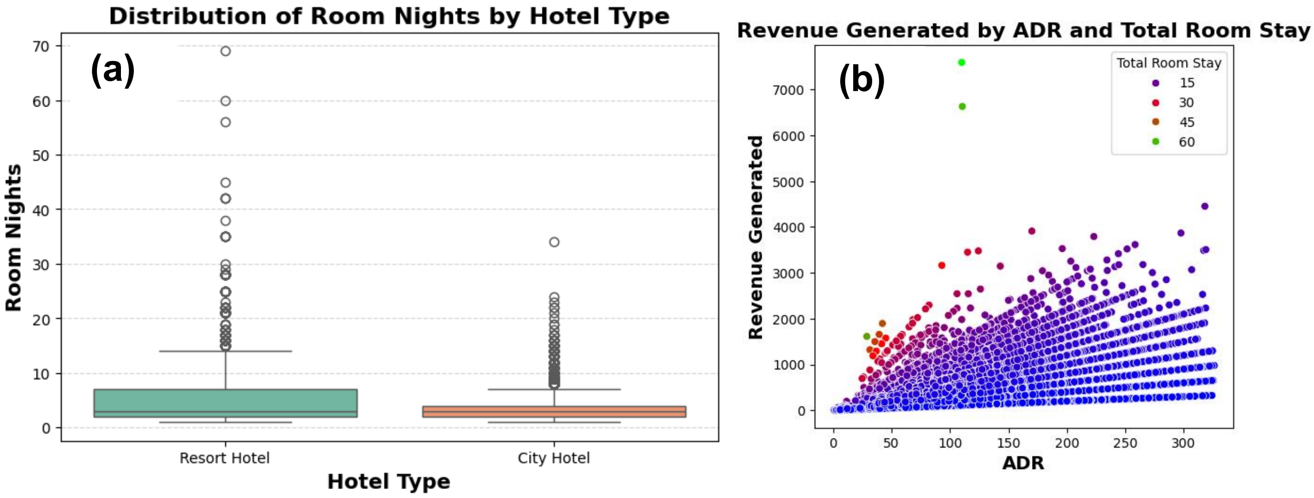

Figure 10: *Distribution of total night stay by* ***(a)*** *hotel type,* ***(b)*** *Total night stay and corresponding adr*

Figure 11 explores the relationship between total night stay, lead time, and revenue across different market segments and customer types. The analysis in **Figure 11a** shows that shorter stays dominate across most booking channels, with most reservations concentrated between one and ten nights. As stay duration increases, the number of bookings declines significantly. This is in alignment with earlier observations that the hotel primarily operates within a short-stay and high-frequency booking environment. The relationship between total night stay and lead time also demonstrates that longer stays are generally associated with earlier booking behaviour. Customers booking extended stays tend to reserve accommodation well in advance, and they often book via OTAs and offline-OT/TA. In contrast, shorter stays are more widely distributed across all lead-time categories and are often linked to more immediate or flexible travel decisions. This further reinforce the earlier observations that planned bookings are closely connected with longer stay durations and leisure-oriented travel behaviour.

The revenue analysis in **Figure 11b** demonstrates a positive relationship between total nights stayed and revenue generation, where longer stays tend to produce significantly higher revenue values. Transient customers dominate the distribution, generating the highest concentration of both room nights and revenue. This support the previous findings that transient two-adult travellers are the hotel's primary source of occupancy and income. Transient-party customers also contribute meaningfully, while contract and group customers remain comparatively limited.

Noticeably, several high-revenue outliers are associated with extended stays which indicate that a smaller number of customers generate exceptionally high booking values through long-duration reservations, premium room selections, or combined booking activities. These findings align with earlier analyses showing that although the hotel relies heavily on high-volume short stays, longer-stay customers represent an important source of higher-value revenue.

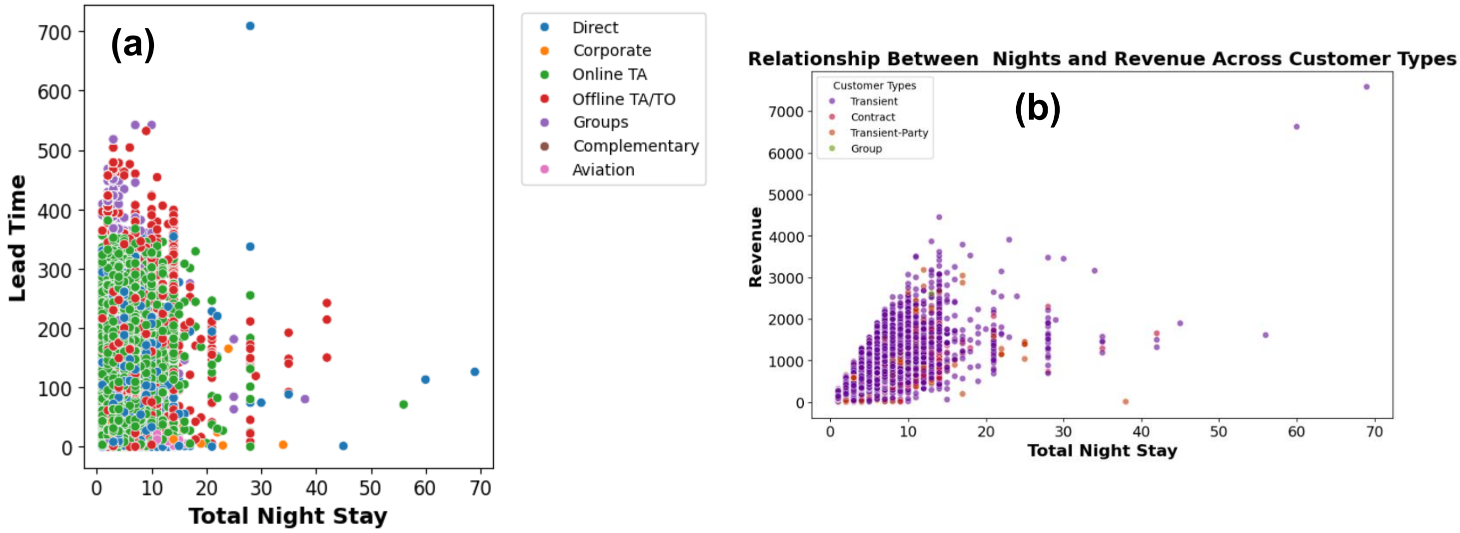

***Figure 11:** Distribution of total night stay across **(a)** market segments and lead time **(b)** customer type and their generated revenue*

4.5 Seasonal Trends

Figure 12 revealed the distribution of bookings across months and seasons. As shown in **Figure 12a**, the highest volume of arrivals occurs in July and August, which corresponds with the summer holiday period. This peak aligns with typical vacation patterns where demand for hotel stays increases significantly, as further highlighted by the seasonal breakdown in **Figure 12b**. In addition to the summer peak, moderate booking volumes are observed during the spring and autumn seasons, indicating steady demand outside the peak months.

In contrast, the lowest level of bookings occurs during the winter months, particularly between November and January. This decline may be attributed to a combination of factors, including the festive period when individuals tend to stay with family, as well as colder weather conditions, which generally reduce leisure travel. The seasonal booking pattern shows a clear distinction between peak (summer) and off-peak (winter) periods.

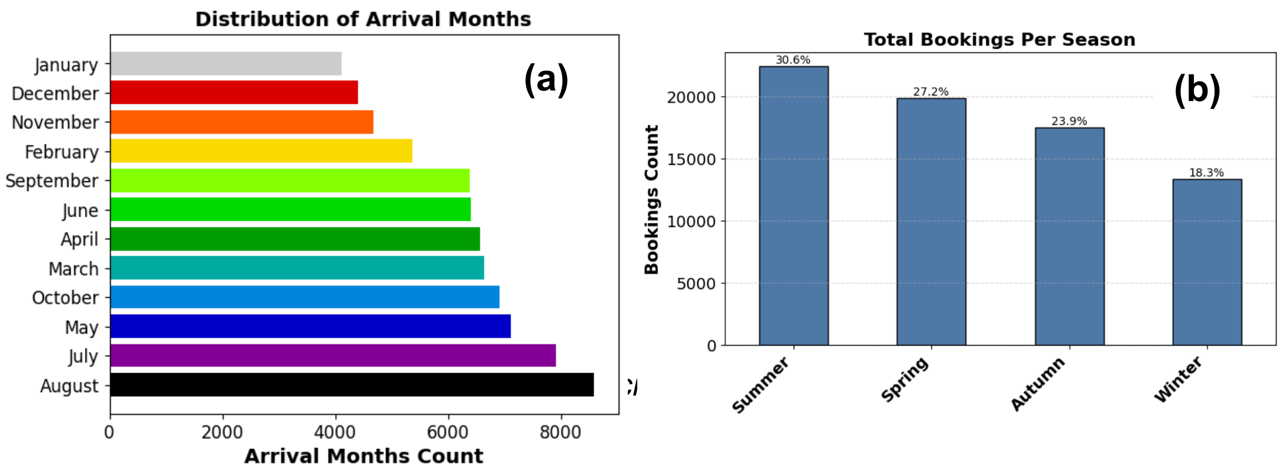

Figure 13 presents the yearly and monthly revenue trends. The yearly analysis (**Figure 13a**) shows a significant increase in revenue from 2015 to 2016, followed by a slight decline in 2017. The decline in 2017 may have been caused by a significant event, or it could indicate that the data for that year is incomplete in the dataset. Despite the decrease in 2017, revenue levels remain substantially higher than in 2015, which mean there is an overall growth trend across the observed period.

The monthly revenue distribution (**Figure 13b)** reveals strong seasonality, with revenue steadily increasing from January and peaking sharply during July and August before declining toward the end of the year. August records the highest revenue generation, closely followed by July. This is line with earlier observation that summer period represents the peak travel season with the highest booking activity and occupancy levels. In contrast, the lowest revenue levels are observed during January, November, and December. This reflects weaker seasonal demand during colder months.

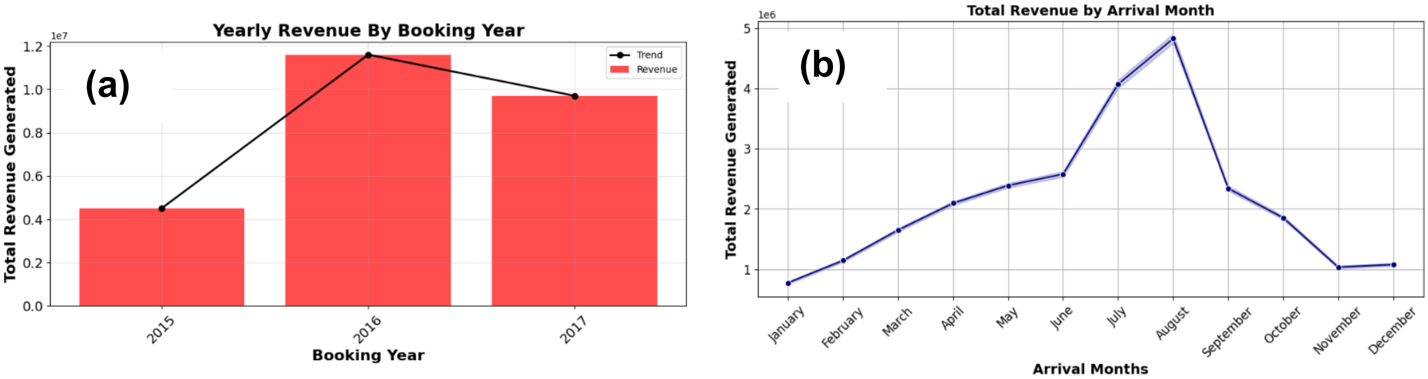

***Figure 13:** Distribution of Total revenue generated across **(a)** booking year, and **(b)** booking months*

Figure 14 examines the relationship between seasonal booking behaviour, lead time, and total night stays. The analysis in **Figure 14a** shows that summer records the highest proportion of early bookings, with early reservations accounting for a substantial share of total bookings during the season. This suggests that customers are more likely to plan summer travel well in advance, reflecting the strong demand associated with peak holiday periods and leisure travel. In contrast, Winter shows a relatively higher proportion of last-minute and short-term bookings, indicating more spontaneous and flexible travel behaviour during lower-demand periods. Spring and Autumn display a more balanced distribution across booking categories, combining both planned and shorter-notice reservations.

The total night stay analysis in **Figure 14b** further confirmed the dominance of summer bookings, which contributes the largest share of overall room nights at approximately 34.5%, followed by Spring (26.4%) and Autumn (23.3%). Winter contributes the smallest proportion of total night stays at 15.8%, confirming earlier findings that travel demand and occupancy levels decline significantly during colder months.

These findings connect with previous analyses showing that summer period generates the highest booking volumes and revenue performance, particularly among transient two-adult travellers and leisure-oriented customers. The concentration of longer stays and early bookings during summer also reveals the importance of seasonal planning behaviour in driving hotel occupancy and profitability.

This finding confirms that seasonal demand patterns significantly influence booking timing and stay duration, with peak travel seasons generating both longer stays and earlier reservations. This creates opportunities for hotels to optimise seasonal pricing, promote advance-booking incentives, and develop off-season strategies aimed at improving occupancy during lower-demand periods.

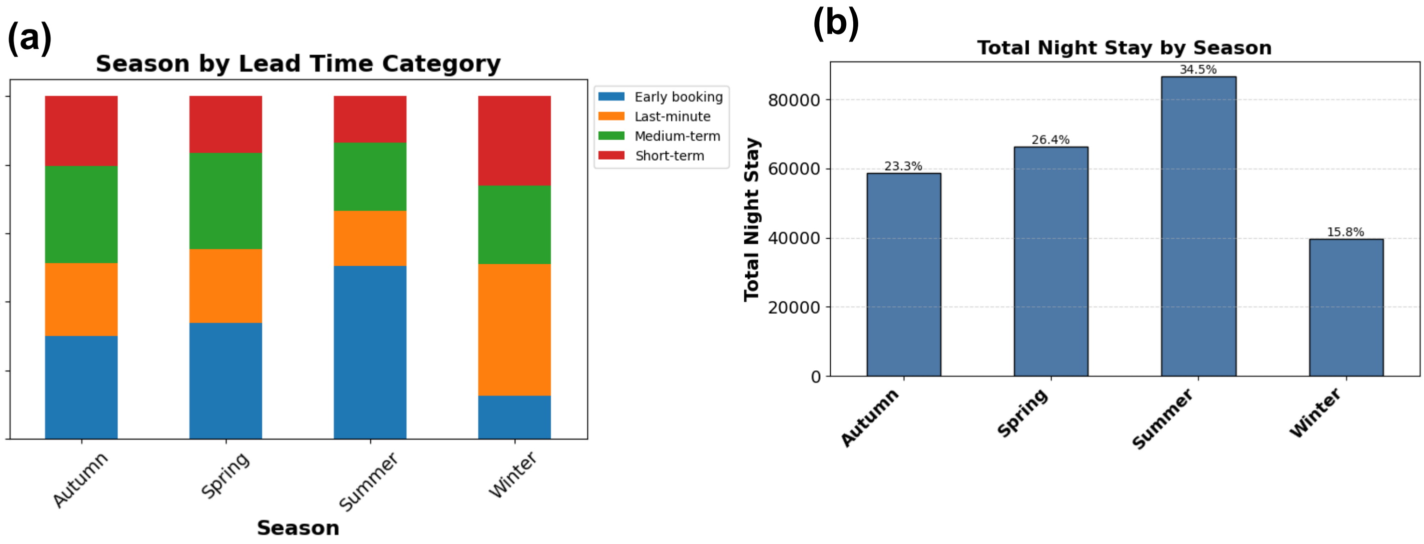

Figure 14: *Distribution of season by **(a)** lead time category, and **(b)** Total night stay*

4.6 Geographical Insights

Figure 15 presents the distribution of bookings and revenue generated by the top 10 customer countries. Portugal contributes the largest share of both bookings and revenue, accounting for approximately 32.7% of total bookings (**Figure 15a)** and 26.1% of total revenue (**Figure 15b**). This indicates a strong dependence on domestic travel demand within the dataset. The United Kingdom and France also represent major international markets, followed by Spain and Germany. This indicates that most bookings originate from European countries, suggesting that the hotels are likely located within Europe. Together, these countries contribute substantial proportions of both bookings and total revenue. This signals a strong market concentration within a relatively small number of key source European countries

A comparison between booking share and revenue contribution further shows that some countries generate proportionally higher revenue relative to their booking volume, indicating stronger spending behaviour and potentially longer stays or higher ADR values.

This finding reveals the hotel's strong reliance on European travel markets, particularly domestic and neighbouring countries. These findings create opportunities for hotels to strengthen targeted international marketing strategies, develop country-specific promotional packages, and expand high-value tourism segments to further diversify revenue sources and reduce dependence on a limited number of markets.

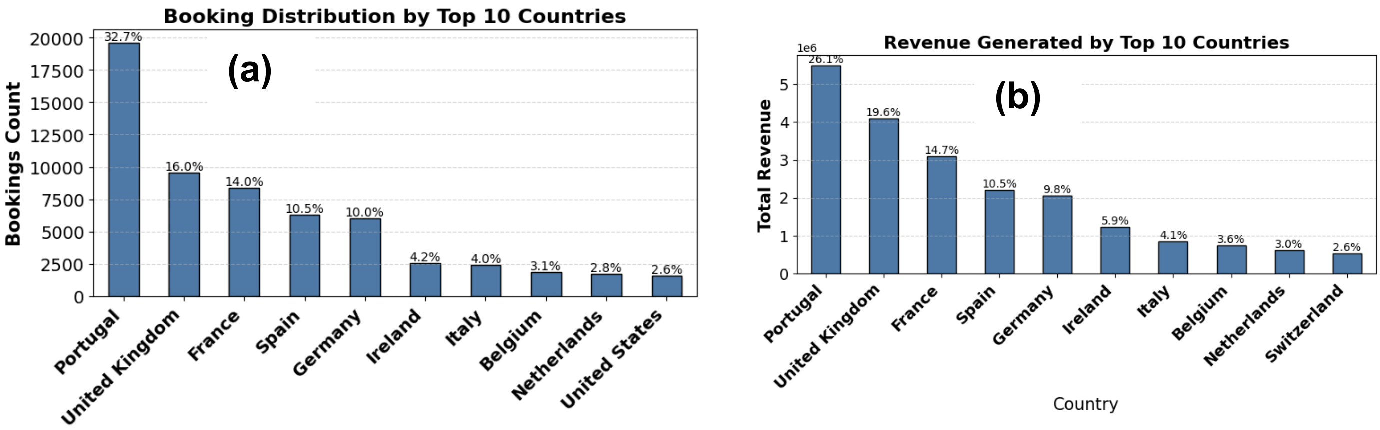

Figure 15: *Booking distribution across (a) Top 10 countries, (b) and their corresponding generated revenue*

The seasonal booking patterns and lead-time behaviour across the top customer countries is presented in **Figure 16**. The seasonal heatmap in **Figure 16a** shows that most countries record stronger booking activity during summer. Portugal consistently records the highest booking volumes across all seasons, confirming its dominance as the hotel's primary market source.

The United Kingdom, France, and Spain also demonstrate strong seasonal demand, particularly during spring and summer period. In contrast, countries such as Belgium, Ireland, and the Netherlands contribute comparatively lower booking volumes across all seasons. Winter generally records lower booking activity across most countries, supporting earlier observations of reduced occupancy and revenue during off-peak periods.

The lead-time analysis (**Figure 16b**) further shows that early bookings dominate across most countries, especially for Portugal and the United Kingdom, indicating that customers from these markets tend to plan trips well in advance. Portugal also records a notable concentration of last-minute bookings, and this suggests a mix of both planned and spontaneous domestic travel behaviour. Meanwhile, medium-term and short-term bookings remain more evenly distributed across other European markets.

These findings align with previous analyses showing that planned bookings, transient travellers, and summer travel periods drive most of the hotel demand and revenue generation. The results also suggest that international markets exhibit different booking behaviours, which creates opportunities for hotels to implement country-specific marketing strategies, seasonal promotions, and targeted advance-booking campaigns tailored to different travel patterns.

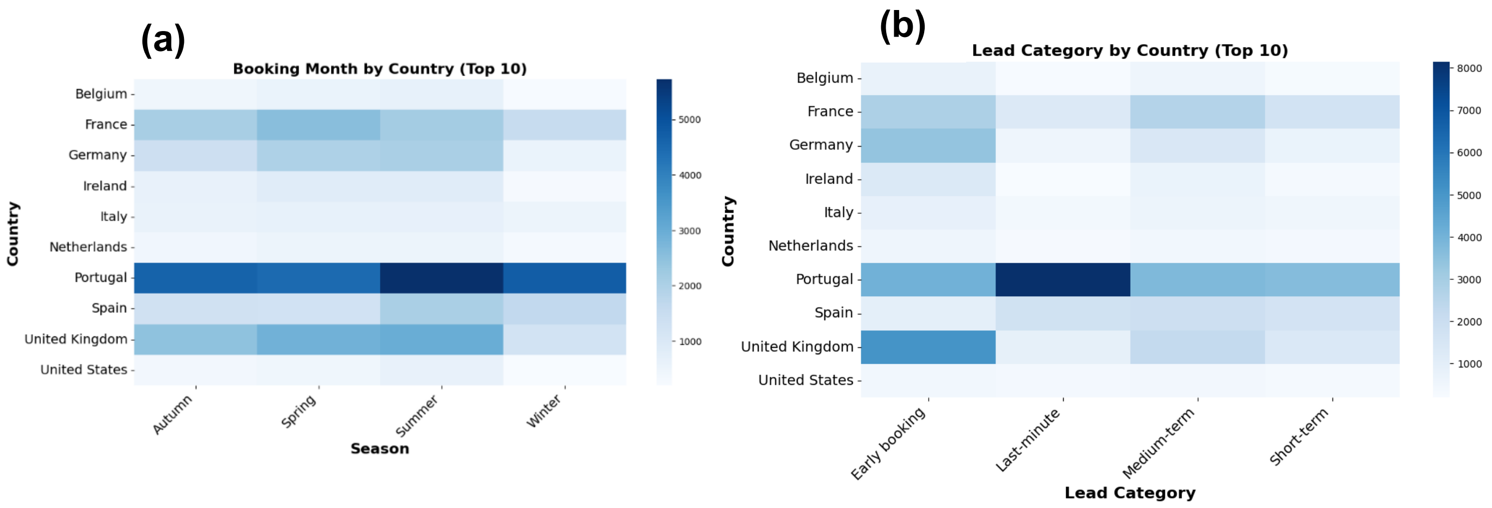

Figure 16: *Heat map showing the distribution of bookings from countries across **(a)** seasons, and **(b)** lead time categories.*

5.0 SQL Analysis

SQL (Structured Query Language) was further employed to analyse and get insights into the cleaned Hotel Booking Demand dataset. While Python was mainly used for data cleaning, preprocessing, visualisation, SQL was applied to further answer key business questions related to booking behaviour, customer trends, average revenue metrics, and hotel operations. Using SQL enabled efficient analysis provided more business-focused insights that support strategic decision-making within the hospitality industry.

5.1 Business Analysis and Key Findings

The SQL analysis revealed several important patterns relating to hotel demand, customer behaviour, pricing, and average revenue generation, and the findings strongly aligned with the earlier exploratory data analysis (EDA). The results showed that City Hotels received substantially more bookings (44,967) than Resort Hotels (28,200). Seasonality also emerged as a major factor influencing booking activity and revenue performance. August recorded the highest number of bookings (8,406) and generated the highest total revenue (22.1 million), confirming that summer represents the peak travel season within the dataset. In contrast, January recorded the lowest revenue levels. This proves that the demand is weak during off-peak travel periods.

From the average revenue perspective, room Type E generated the highest average revenue (3557.94), while Room Types G and H recorded the highest ADR values. This indicates that premium room categories achieve stronger pricing performance and generate higher revenue per booking. Customer segmentation analysis also showed a clear difference in spending behaviour as contract customers generated the highest average revenue (4855.14), followed by transient customers. This indicates that long-term and higher-value bookings contribute significantly to overall profitability. Similarly, certain booking channels attract more valuable customers than others. For instance, the Aviation segment generated the highest average revenue among booking channels, followed by Offline TA/TO and Direct bookings.

In support of the EDA insights, the geographical analysis shows that Portugal, the United Kingdom, and France contributed the highest booking volumes. This further confirms that the hotel demand is strongly concentrated within key European markets. In addition, lead-time analysis showed that Transient-Party and Contract customers tend to book significantly earlier than other customer groups.

Although the SQL analysis focused primarily on average revenue metrics, while the EDA provided broader insights into total revenue patterns, both approaches produced closely aligned findings. The results consistently confirm that hotel revenue performance is strongly influenced by seasonality, room type, customer segment, booking channel, and stay behaviour. The analysis also emphasises the importance of transient travellers, premium room categories, summer demand, and longer stays as key drivers of hotel profitability.

6 Data Pre-Processing for Modelling

The heatmap in **Figure 17a** presents the correlation matrix between independent variables, which is important for detecting multicollinearity in linear regression modelling. The analysis reveals a very strong positive correlation between room_nights and stays_in_week_nights (0.94), as well as between room_nights and stays_in_weekend_nights (0.76). Similarly, stays_in_week_nights andstays_in_week end_nights are also strongly correlated (0.50). These high correlations indicate that the variables capture closely related information about customer stay duration. From a preprocessing perspective, this is important because multicollinearity can negatively affect linear regression performance by inflating coefficient estimates and reducing model interpretability. As a result, highly correlated variables may need to be removed, combined, or carefully selected before training the model. For example, room_nights could serve as a consolidated feature representing total stay duration rather than including multiple overlapping stay variables simultaneously.

The matrix also shows that ADR has a moderate positive relationship with adults (0.32), children (0.34), and revenue (0.54), suggesting that pricing is partly influenced by guest composition and overall booking value. Meanwhile, most remaining variables exhibit relatively weak correlations with each other, indicating lower risks of severe multicollinearity across the wider dataset.

The heatmap correlation between individual features and revenue is shown in **Figure 17b**. The results show that room_nights has the strongest positive correlation with revenue (0.76), followed closely by stays_in_week_nights (0.71), stays_in_weekend_nights (0.58), and ADR (0.54). These variables are therefore identified as the strongest revenue drivers within the dataset. This further confirmed that longer stays and higher ADR values contribute significantly to total revenue generation.

Guest-related variables such as total_guests (0.32), adults (0.27), and children (0.19) show weaker but still positive relationships with revenue, meaning that larger group type such as families may contribute to increased spending and longer stays. Lead_time (0.21) and total_of_special_requests (0.18) also demonstrate mild positive correlations, which means that customers who book earlier or request additional services may generate slightly higher revenue. In contrast, variables such as previous_cancellations, previous_bookings_not_canceled, and room_change show weak negative correlations with revenue. The room_change variable records the strongest negative relationship (-0.16), suggesting that booking modifications or room reallocations may slightly reduce revenue performance due to operational inefficiencies. Some variables, including arrival_date_day_of_month and booking_id, show almost no correlation with revenue and may contribute limited predictive value to the regression model.

The correlation analysis provides an important foundation for feature selection and preprocessing for the linear regression modelling process.

6.1 Limitations of heatmap

Although correlation heatmaps are useful for identifying relationships between variables, relying on them alone has several limitations. First, heatmaps only capture linear relationships and may fail to detect non-linear patterns that still influence revenue. Second, correlation does not imply causation, meaning, strongly correlated variables may not necessarily be true revenue drivers.

Heatmaps also identify multicollinearity but do not determine which variables should be removed or retained. In addition, they cannot capture complex interactions between features, where variables may become important only when combined with others. The results may also be distorted by outliers and skewed data distributions, which are common in hotel booking datasets.

Furthermore, heatmaps provide only exploratory insights and do not measure actual model performance or predictive accuracy. Therefore, additional techniques feature engineering, and model evaluation are required to build a reliable linear regression model for revenue prediction.

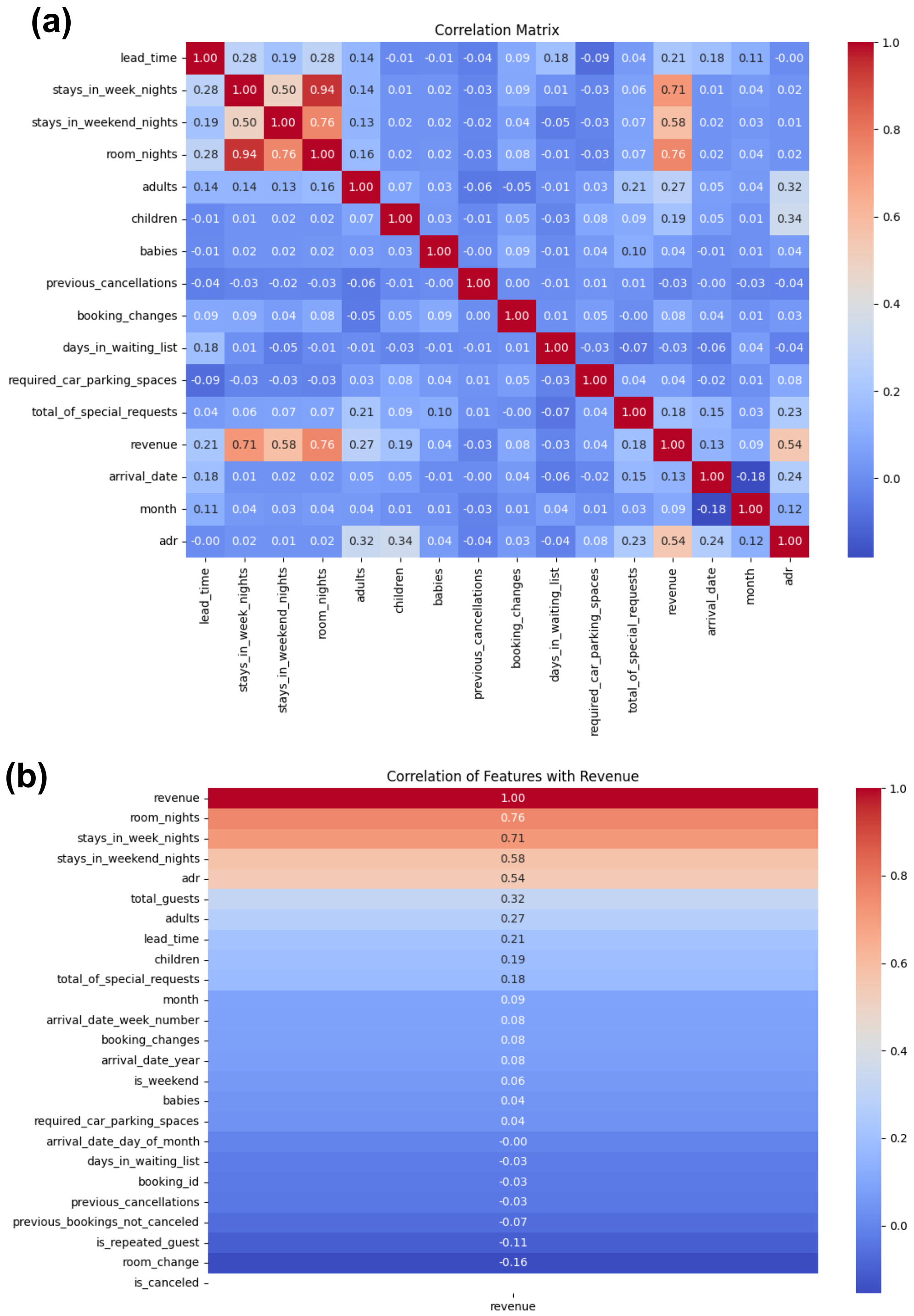

Figure 17: *The heat maps showing **(a)** the correlation relationship between numerical features **(b)** the correlation relationship between revenue and other numerical features.*

6.2 Feature Engineering for Linear Regression Model

Based on the heatmap analysis, redundant variables that were unlikely to contribute significantly to revenue prediction were removed to reduce feature redundancy within the model. In addition, variables that contained highly similar information and can negatively affect the performance and interpretability of linear regression models were dropped to minimise multicollinearity. After removing the redundant features, the dataset contained 73,167 observations and 26 remaining columns. Additionally, a new feature was created to identify the top 10 countries based on booking volume and revenue contribution, while the remaining countries were categorised as 'Others'. This was done to prevent excessive dummy variable encoding in the model.

Machine learning algorithms such as linear regression cannot directly interpret categorical variables unless they are transformed into numerical form. Considering this, categorical features were encoded into binary variables using the one-hot encoding technique, which is suitable for handling nominal and categorical data. The encoded features were initially stored in Boolean format and were subsequently converted into integer format for model compatibility. Following the encoding process, the final dataset consisted of 73,167 observations and 54 columns available for model training.

7 Model development

Identifying the true drivers of total booking value is quite tricky. At a basic level, revenue can be treated as the target variable (label), while variables such as ADR and total night stay are part of the predictor features. However, this approach will likely introduce a significant risk of data leakage into the model because revenue was directly calculated from ADR and total room nights using **Equation 1**. As a result, the model may simply relearn the revenue calculation formula rather than identify the underlying factors that genuinely influence booking value. In addition, there is a methodological concern regarding the suitability of linear regression for this task. Linear regression models relationships as a weighted linear combination of features, as represented in **Equation 2**:

$$W_i = X_1(Feature\ 1) + \ X_2(Feature\ 2) + \cdots + X_n(Feature\ n) + \ W_o \qquad \textbf{2}$$

Where:

- W_i represents the predicted output (label),
- X_1-X_n represent the coefficients assigned to each feature,
- W_o represents the intercept or bias term.

The coefficients and bias are estimated during model training through optimisation techniques such as gradient descent, which aim to minimise prediction error. However, revenue generation process itself is fundamentally multiplicative, since revenue is derived primarily from ADR multiplied by total room nights. This may create a mismatch between the underlying revenue calculation and the additive structure assumed by linear regression which may affect model interpretation and predictive reliability. To address this issue, two modelling approaches were developed and compared. In Model 1, revenue was retained as the target variable to examine the direct drivers of total booking value, despite the acknowledged risk of data leakage. In Model 2, revenue was removed from the dataset, and ADR was used as the target variable instead. This second approach was designed to identify the factors influencing booking value independently of the derived revenue calculation.

By comparing the results of both models, a clearer understanding of the true revenue drivers can be obtained. The modelling results are also interpreted alongside the earlier exploratory data analysis (EDA) findings to provide a more reliable and comprehensive explanation of the factors influencing hotel booking value and revenue generation.

In both models, the dataset was divided into training and testing sets (80/20 data split), where 80% of the dataset was used for model training and the remaining 20% was reserved for testing and evaluation. This approach ensures that the model is trained on a sufficiently large portion of the data while retaining an independent subset to reliably assess its generalisation performance and prevent overfitting.

7.1 Model 1 - What drives total booking value?

In this modelling approach, revenue was used as the target variable (y), while the remaining variables were used as predictor features (X). After fitting the linear regression model, Top 5 predictions were generated and compared with the actual revenue values, as presented in **Table 2**. The results show relatively small prediction errors for larger revenue values, suggesting that the model is likely capturing the main underlying relationships within the dataset. However, larger relative errors are observed for smaller revenue values, indicating reduced model stability and prediction accuracy at lower revenue levels. The prediction pattern reflects a combination of both overprediction and underprediction, showing that the model is not perfectly reconstructing the original revenue-generating relationship despite the presence of data leakage.

This limitation arises because linear regression models relationships using an additive structure, whereas revenue in this dataset is fundamentally generated through a multiplicative relationship between ADR and total room nights. As a result, the model attempts to approximate a multiplicative process using a linear additive framework, which introduces prediction inconsistencies.

Although the model demonstrates reasonably low prediction errors in some cases, its overall validity is compromised due to data leakage. Since ADR and total room nights are direct components used to calculate revenue, including them as predictor variables means the model is effectively relearning an existing mathematical formula rather than discovering independent predictive patterns. Consequently, the model's apparent performance may be misleading and should not be interpreted as evidence of strong real-world predictive capability or reliable forecasting performance.

Table 2: *Comparison of actual revenue and predicted revenue with corresponding error*

Actual revenue	Predicted revenue	Error	|Error|
255.0	257.9	-2.9	2.9
949.5	943.8	5.7	5.7
75.0	110.2	-35.19	35.2
143.4	99.9	43.5	43
225.0	190.4	34.8	34.6

The model achieved an R^2 score of 0.86, indicating that 86% of the variability in revenue is explained by the predicted values (y). Additionally, the root mean square error (RMSE) of 120.4 suggests that predictions deviate from the actual revenue by approximately 120 units on average. While these metrics suggest strong predictive performance, they are likely inflated due to data leakage, as ADR and total nights which are key components used to calculate revenue were included as input features. This likely means predicted values (y) partially reconstruct the target rather than learning independent patterns.

The regression coefficients in **Figure 18** represent the marginal contribution of each feature to the predicted revenue (y), holding other variables constant. In other words, each coefficient indicates how much the predicted revenue increases when that feature is present or increases by one unit. The regression coefficients indicate that several key drivers of revenue identified in the exploratory analysis such as room nights, customer type (Transient and Transient-Party), country (Portugal) and seasonality (Summer) are positively associated with predicted revenue (y), suggesting that the model captures some meaningful business patterns. Although group bookings were initially linked to lower total revenue in the EDA, the model reveals that they can still drive higher total revenue, likely due to longer stays or larger booking sizes. Also, the model predicted that winter season, repeated guest, and non-refund deposit type could drive higher revenue. However, the presence of room nights as a feature introduces data leakage, as revenue is directly derived from length of stay, leading to an inflated R^2 score of 0.86. Also, the positive coefficient for complementary bookings is counterintuitive, as these bookings typically generate minimal revenue. This discrepancy is likely due to dependency effects, where complementary bookings are associated with other features such as longer stays, causing the model to attribute indirect revenue effects to them. In general, the model captures the main revenue drivers effectively, but certain coefficients should be interpreted with caution due to underlying feature relationships.

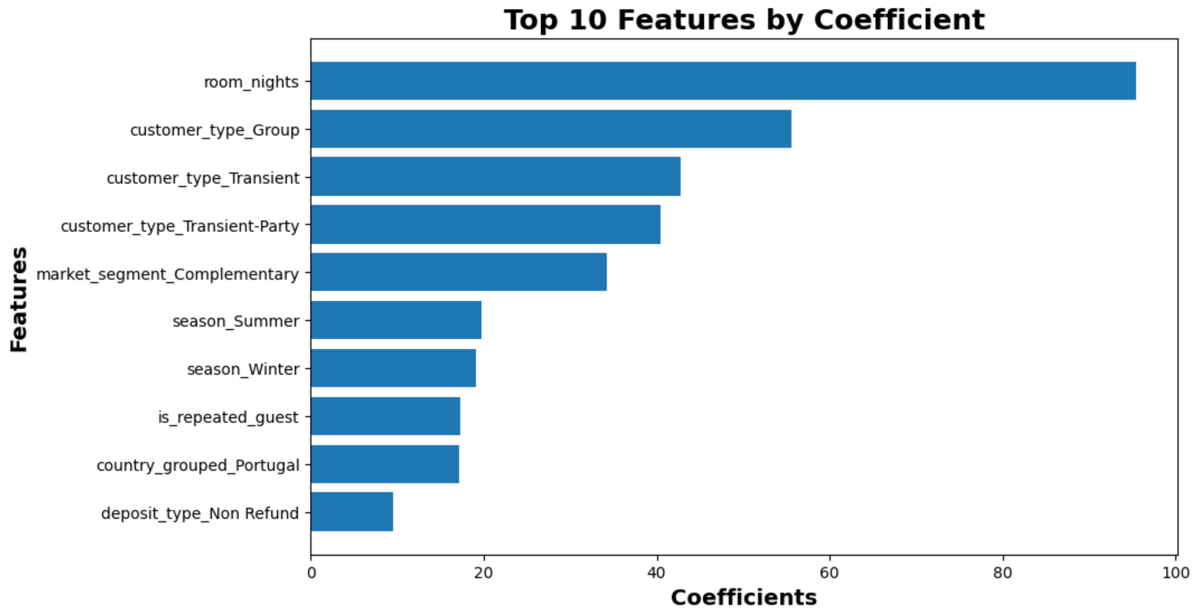

Figure 18: *Regression coefficients showing how importance the features are to the revenue.*

7.2 Model 2 - What influences booking value?

Model 2 achieved an R^2 score of 0.52, meaning that approximately 52% of the variability in pricing is explained by the model. Compared to the revenue model, this reflects a more moderate level of predictive performance given the inherent complexity of pricing decisions in the hospitality sector. The RMSE of 31.94 suggests that predicted ADR values deviate from actual values by an average of around 32 units, which is relatively substantial given the scale of ADR. Inspection of Top 5 predictions (y) in **Table 3** shows a mix of accurate estimates and larger errors, indicating that while the model captures general pricing patterns, it struggles with consistency across all cases. Unlike revenue, which is structurally determined by ADR and length of stay, this means that ADR is influenced by more complex and less observable factors such as pricing strategy and demand dynamics.

Table 3*: Comparison of actual ADR and predicted ADR with corresponding errors*

Actual revenue	Predicted revenue	Error	|Error|
85.0	66.61	18.4	18.4
105.5	107.9	-2.46	2.5
75.0	73.5	1.5	1.5
71.7	65.5	6.2	6.2
75.0	92.8	-17.8	17.8

The ADR model coefficients as shown in **Figure 19** indicate that pricing is primarily driven by room characteristics, seasonality, and booking channel. Higher-tier room types such as G, F, and H have the strongest positive impact on predicted ADR (y), confirming that room quality is a key determinant of price. Seasonal effects are also evident, with summer significantly increasing ADR, reflecting higher demand during peak periods in alignment with the EDA findings. Additionally, direct and online travel agent (OTA) bookings are associated with higher ADR, suggesting that these channels attract or maintain higher-value bookings. The positive effect of total guests indicates that larger bookings may involve premium room allocations further increasing price per night. Unlike the revenue model, customer type and country-level effects are not dominant, suggesting that these factors influence total revenue more through booking volume and stay duration rather than pricing. In essence, while the model captures key pricing drivers consistent with the EDA, its moderate R^2 of 0.52 and RMSE of 31.94 reflect the inherent complexity of ADR, which is influenced by dynamic pricing strategies and external factors not fully represented in the dataset.

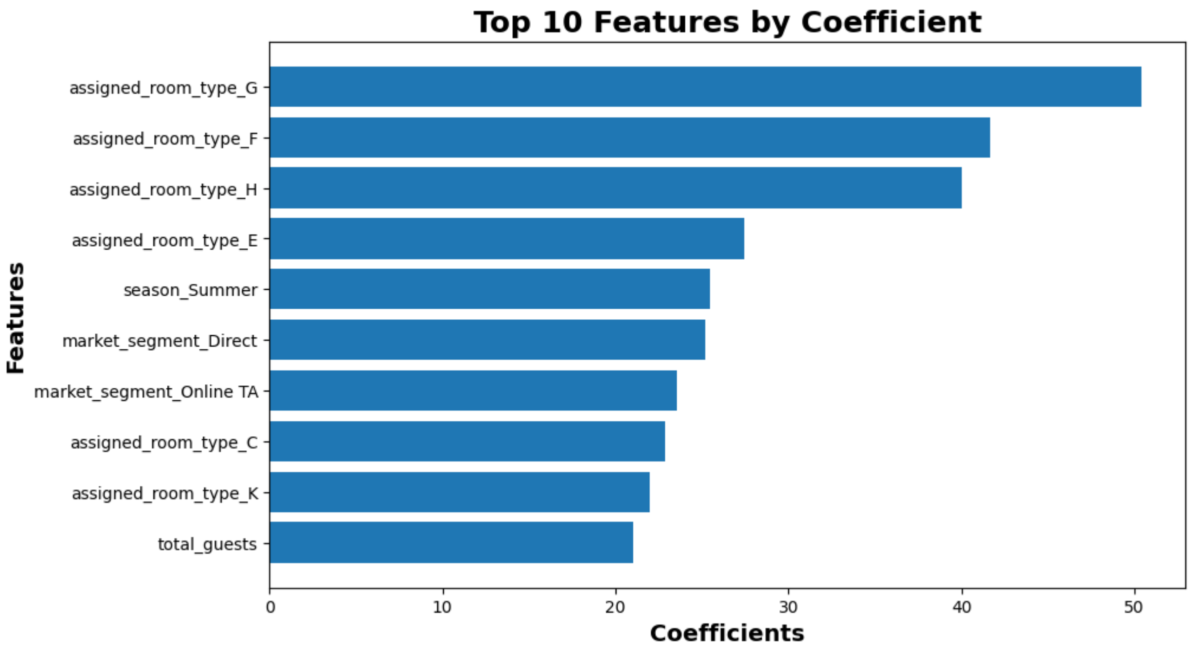

Figure 19: *Regression coefficients showing how importance the features are to the pricing (adr).*

7.3 Comparison of Model 1 and 2

The revenue and ADR models provide complementary insights into the drivers of revenue performance. The revenue model achieved a high R^2 of 0.86, showing its strong predictive capability. The high R^2 score is largely driven by key factors such as room nights, customer type, and seasonality which align well with patterns identified in the exploratory analysis. However, this strong performance is partly influenced by the structural relationship between revenue and length of stay. In contrast, the ADR model, with an R^2 of 0.52 demonstrates more moderate predictive power given the inherent complexity of pricing decisions. Its coefficients show that ADR is primarily determined by room type, booking channel, and seasonal demand. This model provides a more realistic representation of predictive capability, despite its lower performance metrics. While, the revenue model captures volume-driven dynamics, the ADR model provides deeper insight into value and pricing strategy.

8 Business insights and recommendations

The analysis reveals that hotel revenue performance is driven by two major factors: booking volume and pricing strategy. The findings show that room nights, stay duration, and customer booking patterns are the strongest drivers of total revenue, while ADR is more strongly influenced by room type, seasonality, and booking channel behaviour. This highlights the importance of adopting a dual modelling approach, where both Revenue and ADR models are used together to provide complementary business insights. While the Revenue model explains volume-driven performance, the ADR model captures pricing and value differentiation. Using both approaches together enables the business to separate occupancy effects from pricing effects and make more informed operational and strategic decisions.

The results also indicate that revenue growth should not rely solely on increasing room prices. Since longer stays and booking volume contribute significantly to total revenue, hotels should focus on strategies that encourage extended stays and higher occupancy. This may include introducing discounts for longer bookings, bundled accommodation packages, and promotional offers designed to increase room-night generation. The findings further show that transient and transient-party customers remain the hotel's dominant customer segments. As a result, targeted retention and upselling strategies should be developed for these high-performing customer groups.

From a pricing perspective, the ADR model demonstrates that premium room types, summer travel periods, and direct or online booking channels are associated with higher pricing levels. This suggests that pricing strategies should become more dynamic and demand-driven rather than applying uniform pricing across all customer groups and seasons. Hotels should therefore increase prices strategically during peak travel periods, optimise pricing for premium room categories, and strengthen direct booking channels, which are associated with higher ADR and improved profitability.

The analysis also highlights several important market opportunities. Family and group travellers, although underrepresented, demonstrate higher spending behaviour and longer stays, making them valuable growth segments. Hotels can capitalise on this opportunity by offering family-oriented packages, flexible group booking options, and larger accommodation facilities tailored to these customer groups. In addition, Portugal emerged as one of the strongest contributors to bookings and revenue, suggesting the need for stronger market targeting local partnerships and loyalty strategies focused on high-performing geographic markets.

Finally, while linear regression provided a useful baseline for understanding the relationships between hotel booking variables and revenue drivers, the model has important limitations due to its assumption of linear relationships [8]. Hospitality booking behaviour is often influenced by complex and nonlinear interactions involving seasonality, pricing, room type, and booking channels. Therefore, more advanced machine learning techniques such as Random Forest and Gradient Boosting models (e.g., XGBoost or LightGBM) are recommended for future work. These models are better suited for capturing nonlinear relationships and feature interactions, which can improve predictive accuracy and provide more robust support for revenue forecasting, pricing optimisation, and strategic decision-making within the hospitality industry.

9 Conclusion

- The analysis shows that the hotel booking environment is highly flexible, convenience-driven, and strongly influenced by online distribution channels, with customers prioritising accessibility, flexibility, and price comparison when making booking decisions.
- City Hotels generated the highest share of bookings and revenue, largely driven by transient two-adult travellers and short-stay demand, while Resort Hotels were more closely associated with leisure-oriented and longer-stay travel behaviour.
- Hotel demand was strongly seasonal, with summer recording the highest booking volumes, longest stays, and strongest revenue performance. Early bookings also dominated during peak travel periods, indicating the importance of planned travel behaviour.
- The findings confirmed a strong relationship between stay duration, ADR, and revenue generation. Longer stays and higher ADR values contributed significantly to overall booking value and hotel profitability.
- Online Travel Agents (OTAs) remained the dominant booking channel, indicating heavy reliance on intermediary platforms. This reveals the need for hotels to strengthen direct booking channels through loyalty programmes, personalised marketing, and customer retention strategies.
- Family and group travellers, although underrepresented, demonstrated strong spending behaviour and longer stays. This suggests an important opportunity for hospitality growth through targeted packages and larger-group offerings.
- Correlation and modelling analysis identified total night stay, room characteristics, ADR, customer type, and booking behaviour as the strongest drivers of revenue and pricing performance.
- The study demonstrates that effective hotel revenue optimisation requires a combination of dynamic pricing, targeted customer segmentation, seasonal demand management, and strategies that encourage longer stays and higher-value bookings.

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